

Learning theories: Piaget, Vygotsky, Bruner, Skinner

अधिगम के सिद्धांत: पियाजे, वाइगोत्स्की, ब्रूनर, स्किनर

What UPSC asks on this

1. Who-gave-what: ZPD, scaffolding, spiral curriculum, operant conditioning, programmed instruction, schema, object permanence.
2. Piaget's four stages: correct order, age bands, and the hallmark of each (object permanence, egocentrism, conservation, hypothetico-deductive reasoning).
3. Classroom situations: name the operant move (positive or negative reinforcement, punishment, shaping, extinction) or the Piagetian stage a child's answer shows.
4. Contrasts: Piaget vs Vygotsky (who said learning leads development), Bruner vs Ausubel (discovery vs reception), reinforcement vs punishment.
5. Match-the-following across theorists, books, terms and experiments (Skinner box, three-mountain task, Bobo doll, puzzle box).
6. Statement sets with one planted falsehood: "Vygotsky coined scaffolding", "reversibility is preoperational", "Skinner favoured punishment".

The map: three schools, four names

Every learning theory answers one question: does learning come from the environment, from the mind, or from other people? UPSC frames the four

theorists as representatives of three schools, so fix the map first.

- **Behaviourism** (Pavlov, Watson, Thorndike, **Skinner**): learning is a change in observable behaviour produced by stimuli and consequences. The mind is a "black box".
- **Cognitive constructivism** (Piaget, Bruner, Ausubel): learning is a change in mental structures; the learner actively builds knowledge.
- **Social constructivism / sociocultural theory** (**Vygotsky**): knowledge is built first between people, through language and culture, then internalised.

Bruner sits in the cognitive camp but, with Wood and Ross, he also gave the sociocultural camp its most famous word, **scaffolding**. Keep that cross-link in mind.

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`mermaid flowchart TD
  A[Learning theories] --> B[Behaviourism]
  A --> C[Cognitive constructivism]
  A --> D[Social constructivism]
  B --> E[Skinner: operant conditioning]
  C --> F[Piaget: four stages]
  C --> G[Bruner: three modes]
  D --> H[Vygotsky: ZPD and scaffolding]
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Delhi school example for the whole map. In a Sarvodaya Vidyalaya, a Class 6 teacher gives a star for every neat worksheet (Skinner), lets pupils fold paper strips to find that $\frac{1}{2}$ equals $\frac{2}{4}$ (Piaget and Bruner), and pairs a stronger reader with a weaker one in a group formed by reading level (Vygotsky). The three approaches are tools, not rivals.

Skinner: operant conditioning

B. F. Skinner (1904–1990, American) held that behaviour is shaped by its consequences: what is reinforced is repeated, what is not reinforced fades.

He called this **operant conditioning** because the organism *operates* on the environment. Pavlov's **respondent** (classical) conditioning is different: a stimulus triggers an existing reflex.

The precursors UPSC mixes in.

- **Pavlov** (Russian, **Nobel Prize 1904** for work on digestion): classical conditioning with dogs; unconditioned stimulus (food), conditioned stimulus (bell), conditioned response (salivation).
- **Thorndike** (American): trial-and-error learning with cats in a **puzzle box**; **Law of Effect** (responses followed by satisfaction are strengthened), **Law of Exercise**, **Law of Readiness**. His S-R bond theory is called **connectionism**. Skinner's reinforcement is the Law of Effect made precise.
- **Watson** (American): founded behaviourism in **1913**; the **Little Albert** experiment (**1920**) conditioned fear of a white rat in an infant.

Skinner's core terms. He studied rats and pigeons in the **Skinner box** (operant conditioning chamber), recording how lever-pressing changed with consequences.

- **Reinforcement** is any consequence that *increases* a behaviour. **Punishment** is any consequence that *decreases* it.

- **Positive** means something is *added*; **negative** means something is *removed*. Positive and negative do not mean good and bad.

Effect on behaviour	Stimulus added	Stimulus removed
Increases (reinforcement)	Positive reinforcement: praise, star, marks	Negative reinforcement: extra homework cancelled
Decreases (punishment)	Positive punishment: scolding, extra work	Negative punishment: loses free period, time-out

The trap UPSC plants every time: **negative reinforcement is not punishment**. It *strengthens* behaviour by taking away something unpleasant. A pupil who finishes classwork on time is excused from the evening worksheet, so finishing on time increases.

- **Primary reinforcers** satisfy biological needs (food). **Secondary (conditioned) reinforcers** acquire value by association: praise, grades, tokens. A **token economy** uses points exchangeable for privileges.
- **Shaping:** reinforcing **successive approximations** of a target behaviour. A shy pupil is praised first for answering from her seat, then for standing to answer, then for speaking from the front.
- **Chaining:** linking separately learned behaviours into a sequence (the steps of a titration).
- **Extinction:** the behaviour fades when reinforcement stops. Ignoring attention-seeking

calling-out is extinction, not punishment.

Spontaneous recovery: the extinguished response briefly reappears after a rest.

- **Generalisation** (responding to similar stimuli) and **discrimination** (responding only to the right stimulus).
- **Premack principle** (David Premack, "Grandma's rule"): a preferred activity reinforces a less preferred one. "Finish the sums, then library time."

Schedules of reinforcement. Continuous reinforcement (every response) builds a behaviour fastest; partial (intermittent) reinforcement makes it last.

- **Fixed ratio (FR):** reward after a set number of responses. High rate, short pause after each reward.
- **Variable ratio (VR):** after an unpredictable number of responses. **Highest, steadiest rate and most resistant to extinction** (slot machines, surprise praise).
- **Fixed interval (FI):** after a set time. Produces a "scallop": little work until just before the reward (studying only before the scheduled test).
- **Variable interval (VI):** after unpredictable times (surprise class tests). Steady, moderate rate.

Memory aid: **Variable Ratio = Vegas Ratio**, the gambler's schedule.

Skinner on punishment. He argued punishment only suppresses behaviour temporarily, teaches nothing

new, and breeds fear and avoidance. He preferred **extinction plus reinforcement of an alternative behaviour**. This sits well with **Section 17 of the RTE Act, 2009**, which bans physical punishment and mental harassment.

Programmed instruction and teaching machines. In his **1954** paper *The Science of Learning and the Art of Teaching* and in *The Technology of Teaching* (**1968**) Skinner applied operant principles to teaching:

- content broken into **small steps** (frames);
- an **active response** by the learner to each frame;
- **immediate feedback** (confirmation works as reinforcement);
- **self-pacing**.

Skinner's programme is **linear**: every learner takes the same path. The **branching (intrinsic) programme**, where a wrong answer sends you to a remedial frame, is **Norman Crowder's**, not Skinner's.

Books for match-the-following. *The Behavior of Organisms* (**1938**), *Walden Two* (1948, a novel), *Science and Human Behavior* (1953), *Verbal Behavior* (1957, attacked by Chomsky), *Beyond Freedom and Dignity* (1971).

Delhi school example. A Class 9 teacher posts a "Star of the Week" board (positive reinforcement with a secondary reinforcer), stops reacting to a pupil's attention-seeking remarks (extinction), and uses surprise five-minute quizzes (variable interval) instead of only the announced unit test (fixed interval). A Vice

Principal observing the class should be able to name each move.

Piaget: stages of cognitive development

Jean Piaget (1896–1980, Swiss) showed that children are not small adults: they think in qualitatively different ways at different ages and build knowledge by acting on the world. He called his field **genetic epistemology** (the origins of knowledge) and used the **clinical interview method**, probing children's wrong answers instead of counting right ones. His position is **cognitive (individual) constructivism**.

How thinking grows: four processes.

- **Schema:** an organised pattern of thought or action, a mental "file" (an infant's sucking schema, a child's "bird" category).
- **Assimilation:** fitting new experience into an existing schema (calling a bat a "bird").
- **Accommodation:** changing the schema to fit the experience (a new category for bats).
- **Equilibration:** the drive to restore balance when new information does not fit (**disequilibrium**). Learning happens at this moment of cognitive conflict. Assimilation and accommodation together are **adaptation**.

The four stages. The *sequence* is invariant and universal; the *ages* are approximate. No stage can be skipped.

``mermaid timeline title Piaget's stages 0–2 yrs : Sensorimotor : Object permanence 2–7 yrs :

Preoperational : Egocentrism, no conservation 7–11 yrs : Concrete operational : Conservation, reversibility 11+ yrs : Formal operational : Abstract, hypothetical thought ``

1. Sensorimotor (birth to about 2). The infant knows the world through senses and movement. Hallmark: **object permanence**, knowing a hidden object still exists (begins around **8 months**, complete by about 18–24 months). Deferred imitation and the first mental representations appear at the end.

2. Preoperational (about 2 to 7). Language and pretend play explode, but thought is not yet logical. The stage is usually split into the **preconceptual (symbolic function) substage (2–4)** and the **intuitive thought substage (4–7)**; "preconceptual" is Piaget's own label, "symbolic function" is the label used in textbooks such as Santrock. Hallmarks:

- **Egocentrism:** cannot take another's viewpoint. Tested by the **three-mountain task** (with Inhelder).
- **Animism:** the sun is "happy", the moon "follows me".
- **Centration:** focuses on one feature (height of a glass) and ignores others (width).
- **Irreversibility:** cannot mentally undo an action.
- **No conservation:** thinks the amount changes when the shape changes.

3. Concrete operational (about 7 to 11). Logical thought about concrete, present objects. Hallmarks: **conservation** (number first, about 6–7; mass and

liquid 7–8; weight 9–10; volume last, 11–12), **reversibility**, **decentration**, **seriation** (ordering sticks by length), **classification** and **class inclusion** (more flowers than roses), **transitivity** ($A > B$ and $B > C$, so $A > C$). Cannot yet reason about purely hypothetical statements.

4. Formal operational (about 11–12 onward).

Abstract, hypothetico-deductive reasoning; systematic testing of hypotheses (the **pendulum task**); proportion, probability and algebra; thinking about thinking; adolescent idealism. Piaget later accepted that many adults do not use formal operations in every domain.

Memory aid: "**Some People Can Fly**" for Sensorimotor, Preoperational, Concrete, Formal, with boundaries at **2, 7, 11**.

Piaget on morality and play. In *The Moral Judgment of the Child* (1932) he described **heteronomous morality** (rules are fixed and sacred; acts judged by consequences; **immanent justice**, misfortune is punishment for breaking a rule) giving way around age 10 to **autonomous morality** (rules are agreements; intentions matter). Kohlberg built his six stages on this. Play follows the stages too: **practice play**, then **symbolic play**, then **games with rules**.

What Piaget means for teaching.

- **Readiness:** teach what the stage allows; a preoperational child cannot be lectured into conservation.

- **Active discovery** with concrete materials. Piaget's own title: *To Understand Is to Invent*.
- Create **cognitive conflict** and let peers argue; disequilibrium drives learning.
- **Errors are informative**: they reveal the child's current schema.
- The teacher is a **facilitator**, not a transmitter.

NEP 2020 link. NEP 2020 (**para 4.1**) reconfigures schooling into **5+3+3+4** stages "corresponding to the age ranges of **3–8, 8–11, 11–14 and 14–18**", chosen to match developmental stages. Read against Piaget: Foundational (3–8) is largely preoperational, so NEP prescribes **play- and activity-based** learning; Preparatory (8–11) is concrete operational, so **discovery and activity** with light textbooks; Middle (11–14) begins formal operations, so **experiential learning** and abstract concepts; Secondary (14–18) assumes formal operations, so **critical thinking** and student choice. **NCF 2005** had already placed the "child as constructor of knowledge" at the centre.

Criticisms (asked as "which statement is incorrect").

Piaget **underestimated** young children (McGarrigle and Donaldson's "naughty teddy" study found conservation earlier when the change looked accidental) and **overestimated** adolescents; development is more **gradual and domain-specific** than his stages imply (**horizontal décalage**: conservation of number years before volume); he **neglected culture, language and instruction**, the gap Vygotsky filled.

Delhi school example. In a Class 3 maths lesson the teacher pours the same water into a tall bottle and a wide one and asks "which has more?" Children who say "same, you just poured it" have conservation. Those who point to the tall bottle are still centring on height. The teacher gives the second group more hands-on pouring, not more worksheets.

Vygotsky: learning is social first

Lev Vygotsky (1896–1934, Soviet psychologist, died of tuberculosis at 37) argued that higher mental functions originate in social interaction and are carried by cultural tools, above all language. His work reached the West through *Thought and Language* (1934, English 1962) and the edited volume *Mind in Society* (1978). The theory is called **sociocultural or social constructivist**.

The general law. Every higher function appears **twice**: first **between people** (interpsychological), then **inside the child** (intrapsychological). This movement from social to individual is **internalisation**.

Zone of Proximal Development (ZPD). The distance between what a child can do **independently** (actual development) and what she can do **with guidance from an adult or a more capable peer** (potential development). Teaching must aim *inside* this zone: too low is boring, too high is impossible. Vygotsky's line: "the only good learning is that which is **in advance of development**." So for Vygotsky **learning leads development**; for Piaget development comes first and sets the limit.

Scaffolding and the MKO. The helper is the **More Knowledgeable Other (MKO)**: teacher, parent, elder sibling, peer, even a well-designed app. (Vygotsky spoke of the adult or "more capable peer"; "MKO" is a label added by later writers.) The temporary, adjustable help the MKO gives is **scaffolding**: modelling, hints, prompts, splitting the task, thinking aloud. As competence grows the support is **faded** and removed. **The word "scaffolding" comes from Wood, Bruner and Ross (1976), not from Vygotsky; UPSC uses this as a false statement.**

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`mermaid flowchart TD
  A[Task inside the ZPD] --> B[More knowledgeable other helps]
  B --> C[Scaffold with hints and prompts]
  C --> D[Child internalises the skill]
  D --> E[Support faded]
  E --> F[Independent performance]`
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Language and thought. Speech develops from **social speech** (to others, up to about 3) to **egocentric or private speech** (talking aloud to oneself, 3–7) to **inner speech** (silent self-talk, from about 7). Piaget saw egocentric speech as immature chatter that disappears; Vygotsky saw it as **self-regulation that goes underground**. A Class 1 child muttering "first the big one, then the small one" while sorting blocks is planning with private speech and should not be hushed.

Other Vygotskian ideas UPSC touches.

- **Cultural tools:** technical tools (abacus, pen) and psychological tools (language, number systems, maps, writing).

- **Play creates a ZPD:** in pretend play a child "behaves beyond his average age".
- **Dynamic assessment:** test what a child can do *with* help, not only alone; the root of "assessment *for* learning".
- **Defectology:** his work on children with disabilities stressed social inclusion and compensation, an early case for inclusive education.

What Vygotsky means for teaching. Cooperative learning, **peer tutoring**, mixed-ability groups, **reciprocal teaching** (Palincsar and Brown: pupils take turns leading summarising, questioning, clarifying and predicting), **cognitive apprenticeship**, teacher as **mediator**, assessment of assisted performance, and teaching in the home language because language is the tool of thought. NEP 2020's mother-tongue emphasis echoes this.

Delhi school example. A teacher groups children by current reading level rather than by class (the approach Delhi used in Mission Buniyaad, one of the programmes replaced in September 2025 by NEEEV, Science of Living and Rashtraneeti). She models a reading strategy, a stronger reader partners a weaker one, and she steps back as fluency grows. That is ZPD, MKO and fading in one lesson.

Bruner: discovery, modes and the spiral

Jerome Bruner (1915–2016, American) led the "cognitive revolution" and asked how teaching should be organised so learners discover the

structure of a subject. He co-founded Harvard's Center for Cognitive Studies (1960) with George Miller. Unlike Piaget's descriptive theory of development, Bruner offers a **prescriptive theory of instruction:** how to teach.

Three modes of representation. Knowledge is held in three ways, usually acquired in this order for any new idea but **not tied to age:**

- **Enactive:** through action (handling a balance, folding a strip).
- **Iconic:** through images (a diagram of the balance).
- **Symbolic:** through language and symbols (the equation).

Memory aid: **E-I-S = Do, See, Say.** Contrast with Piaget: his stages are age-bound and replace one another; Bruner's modes are cumulative and adults use all three.

Discovery learning. In *The Act of Discovery* (1961) Bruner argued that learners who work out a principle themselves gain **intellectual potency**, shift from **extrinsic to intrinsic motivation**, learn the **heuristics of discovery**, and **remember better**. In practice this is **guided discovery:** the teacher structures materials and questions, the learner finds the pattern. Its opposite number is **Ausubel's meaningful reception learning** with **advance organisers**, where the teacher presents well-organised content linked to prior knowledge. Both are cognitive; UPSC contrasts them.

The Process of Education (1960). This short book, from the **1959 Woods Hole conference**, gave four themes:

1. **Structure:** teach the fundamental ideas of a discipline, not isolated facts; structure transfers.
2. **Readiness:** "any subject can be taught effectively in some intellectually honest form to any child at any stage of development." Readiness is not waited for (Piaget) but **taught**.
3. **Intuitive and analytic thinking:** value the intuitive leap, then verify it.
4. **Motives for learning:** curiosity over grades.

Spiral curriculum. Return to the same big ideas at increasing depth and abstraction as the learner matures: fractions as paper strips in Class 3, as points on a number line in Class 5, as rational numbers in Class 8. NCERT textbooks are built this way.

Toward a Theory of Instruction (1966). A theory of instruction must specify **four features**: (1) experiences that create a **predisposition** to learn; (2) how knowledge is **structured** (mode of representation, economy, power); (3) the best **sequence**; (4) the **nature and pacing of rewards and punishments**, moving from extrinsic to intrinsic.

Concept attainment. In *A Study of Thinking* (1956, with Goodnow and Austin) Bruner showed how people form concepts by testing examples and non-examples. The **Concept Attainment Model** in Joyce and Weil's *Models of Teaching* comes from this.

Scaffolding and narrative. With Wood and Ross (1976) he coined **scaffolding**; later he proposed the **Language Acquisition Support System (LASS)**, the adult help that complements Chomsky's innate LAD. His late work (*Actual Minds, Possible Worlds*, 1986) separated **narrative** from **paradigmatic** (logical-scientific) thinking and stressed culture and story in learning.

Delhi school example. A lesson that opens with a story and reflection uses the narrative mode (Delhi's former Happiness Curriculum worked this way; it was replaced in September 2025). A science teacher introducing density lets pupils drop objects in water (enactive), draw what floats and sinks (iconic), then write the formula (symbolic), and revisits density in a later class with Archimedes' principle (spiral).

The contrasts UPSC tests

Most comparison questions turn on four points: what drives development, the order of learning and development, the role of language, and the teacher's job.

Point	Piaget	Vygotsky
Driver	Child's own action; maturation	Social interaction; culture
Order	Development precedes learning	Learning leads development
Language	Product of thought	Tool that shapes thought
Egocentric speech	Immature, fades away	Becomes inner speech
Stages	Four universal stages	No fixed stages
Teacher	Facilitator; provides materials	Mediator; scaffolds in ZPD
Peers	Source of cognitive conflict	More capable peers scaffold

Bruner against both: like Piaget he wants discovery, but like Vygotsky he says culture and instruction, not maturation, set the pace; his **modes are not age stages**; he alone wrote a prescriptive **theory of instruction** and the **spiral curriculum**.

Skinner against all three: no mental structures, no stages, no discovery. Learning is a change in the **frequency of a response** caused by **consequences**; the teacher **arranges contingencies** and programmes content in **small reinforced steps**.

Other names that appear in the same question

- **Bandura:** observational (social) learning; **Bobo doll** experiment (**1961**); attention, retention,

reproduction, motivation; **self-efficacy**. Bridges behaviourism and cognitivism.

- **Köhler: insight learning**; the chimpanzee **Sultan**; Gestalt school.
- **Tolman: latent learning, cognitive maps**; rats in mazes.
- **Gagné: hierarchy of eight types of learning** (signal learning to problem solving); **nine events of instruction**.
- **Ausubel: advance organisers**, meaningful reception learning.
- **Kohlberg: three levels, six stages** of moral reasoning; Heinz dilemma.
- **Erikson: eight psychosocial stages**; identity vs role confusion in adolescence.

Common traps

- **"Vygotsky coined scaffolding."** False. Wood, Bruner and Ross, 1976. Vygotsky gave ZPD.
- **Negative reinforcement treated as punishment.** Negative reinforcement *increases* behaviour by removing something unpleasant.
- **"Reversibility is preoperational."** False. Preoperational thought is *irreversible*; reversibility arrives in the concrete operational stage.
- **Object permanence placed in preoperational.** It belongs to the **sensorimotor** stage.
- **Bruner's modes offered as age stages.** They are modes of representation; adults use all three.

- **"Any subject can be taught..."** is Bruner (*The Process of Education*, 1960), not Piaget or Dewey.
- **Branching programmed instruction** is Crowder; Skinner's is linear.
- **Fixed interval** offered as most resistant to extinction; it is **variable ratio**.
- **"Piaget: learning leads development."** Reverse it: Piaget development first, Vygotsky learning leads.
- **Law of Effect** is Thorndike, not Skinner; **classical conditioning** is Pavlov, not Watson.
- **Three-mountain task** tests egocentrism, not conservation; **Bobo doll** is Bandura, not Skinner.
- **Ages:** Piaget's boundaries are **2, 7, 11**; NEP's are **8, 11, 14**. Do not mix them.

Remember for the exam

1. Skinner: operant conditioning, Skinner box, *The Behavior of Organisms* 1938, teaching-machines paper 1954, linear programmed instruction.
2. Reinforcement increases behaviour; punishment decreases it. Positive adds a stimulus; negative removes one.
3. Variable ratio: highest steady rate, most resistant to extinction. Fixed interval: scallop.
4. Shaping = successive approximations; extinction = withhold reinforcement; Premack = preferred activity as reward.
5. Thorndike: Law of Effect, puzzle box, connectionism. Pavlov: classical conditioning, Nobel 1904. Watson: Little Albert 1920.

6. Piaget: schema, assimilation, accommodation, equilibration; genetic epistemology; clinical method.
7. Piaget stages: Sensorimotor 0–2 (object permanence), Preoperational 2–7 (egocentrism, centration, no conservation), Concrete 7–11 (conservation, reversibility, seriation, class inclusion), Formal 11+ (hypothetico-deductive).
8. Piaget morality: heteronomous (consequences, immanent justice) to autonomous (intentions) around age 10.
9. NEP 2020 para 4.1: 5+3+3+4 = ages 3–8, 8–11, 11–14, 14–18, justified by developmental stages.
10. Vygotsky: ZPD, MKO, internalisation, private speech to inner speech, cultural tools, learning leads development; *Thought and Language* 1934, *Mind in Society* 1978.
11. Scaffolding: Wood, Bruner and Ross 1976; temporary support, faded as competence grows.
12. Bruner: enactive, iconic, symbolic; discovery learning; spiral curriculum; structure; *The Process of Education* 1960; *Toward a Theory of Instruction* 1966 with four features; concept attainment 1956.
13. Ausubel = advance organisers and reception learning, the opposite of Bruner's discovery.
14. Bandura = Bobo doll and self-efficacy; Köhler = insight; Tolman = latent learning; Gagné = eight types, nine events.

Sources

- NCERT, *Psychology* Class XI, Chapter 3 "Human Development" and Chapter 5 "Learning" (current edition).
- IGNOU B.Ed. BES-123 *Learning and Teaching*, Block 1 "Learning: Perspectives and Approaches".
- IGNOU B.Ed. BES-121 *Childhood and Growing Up*, Block 2 "Growing Up: Infancy to Adulthood".
- NIOS D.El.Ed. Course 502, Block 1 "Learning and Teaching Process".
- National Education Policy 2020, Chapter 4, paras 4.1–4.2 (Ministry of Education).
- National Curriculum Framework 2005, Chapter 2 "Learning and Knowledge" (NCERT).
- Right of Children to Free and Compulsory Education Act, 2009, Section 17.
- B. F. Skinner, *The Technology of Teaching* (1968); J. S. Bruner, *The Process of Education* (1960) and *Toward a Theory of Instruction* (1966); L. S. Vygotsky, *Mind in Society* (1978); D. Wood, J. S. Bruner and G. Ross, "The role of tutoring in problem solving" (1976).
- S. K. Mangal, *Advanced Educational Psychology* (PHI), chapters on learning and cognitive development.

Quick facts

- Skinner: operant conditioning; behaviour is controlled by its consequences; Skinner box; linear programmed instruction (1954 paper, *The Technology of Teaching* 1968).

- Reinforcement increases behaviour, punishment decreases it; positive = stimulus added, negative = stimulus removed. Negative reinforcement is not punishment.
- Variable ratio schedule gives the highest, steadiest response rate and is most resistant to extinction; fixed interval gives the scallop.
- Shaping = reinforcing successive approximations; extinction = withholding reinforcement; Premack principle = preferred activity as reward.
- Thorndike: Law of Effect, puzzle box, connectionism. Pavlov: classical conditioning, Nobel 1904. Watson: behaviourism 1913, Little Albert 1920.
- Piaget: schema, assimilation, accommodation, equilibration; genetic epistemology; clinical interview method; cognitive constructivism.
- Piaget's stages: Sensorimotor 0-2 (object permanence), Preoperational 2-7 (egocentrism, centration, irreversibility, no conservation), Concrete operational 7-11 (conservation, reversibility, seriation, class inclusion), Formal operational 11+ (hypothetico-deductive, abstract).
- Piaget's morality (1932): heteronomous (consequences, immanent justice) to autonomous (intentions) around age 10; Kohlberg built on it.

- NEP 2020 para 4.1: 5+3+3+4 stages correspond to ages 3-8, 8-11, 11-14 and 14-18, chosen to match developmental stages.
- Vygotsky: sociocultural theory; ZPD = gap between independent and assisted performance; MKO; internalisation; learning leads development; Thought and Language 1934, Mind in Society 1978.
- Scaffolding was coined by Wood, Bruner and Ross (1976), not Vygotsky; support is temporary and faded as competence grows.
- Private (egocentric) speech: Piaget said it fades as immaturity; Vygotsky said it becomes inner speech and serves self-regulation.
- Bruner: enactive, iconic, symbolic modes (Do, See, Say); discovery learning (1961); spiral curriculum; structure; The Process of Education 1960; Toward a Theory of Instruction 1966 with four features.
- Bruner's readiness quote: any subject can be taught effectively in some intellectually honest form to any child at any stage of development.
- Ausubel = advance organisers and meaningful reception learning (opposite of Bruner's discovery); Bandura = Bobo doll, observational learning, self-efficacy; Kohler = insight; Tolman = latent learning.